

What is Motion?

- **Motion** is the change in the position of an object with respect to time and a reference point.
- An object is said to be in motion if its position changes relative to its surroundings.

Example: A passenger sitting inside a moving bus appears at rest relative to the bus but is in **motion** relative to the road.

Key Elements of Motion



OBJECT

– the thing that moves.



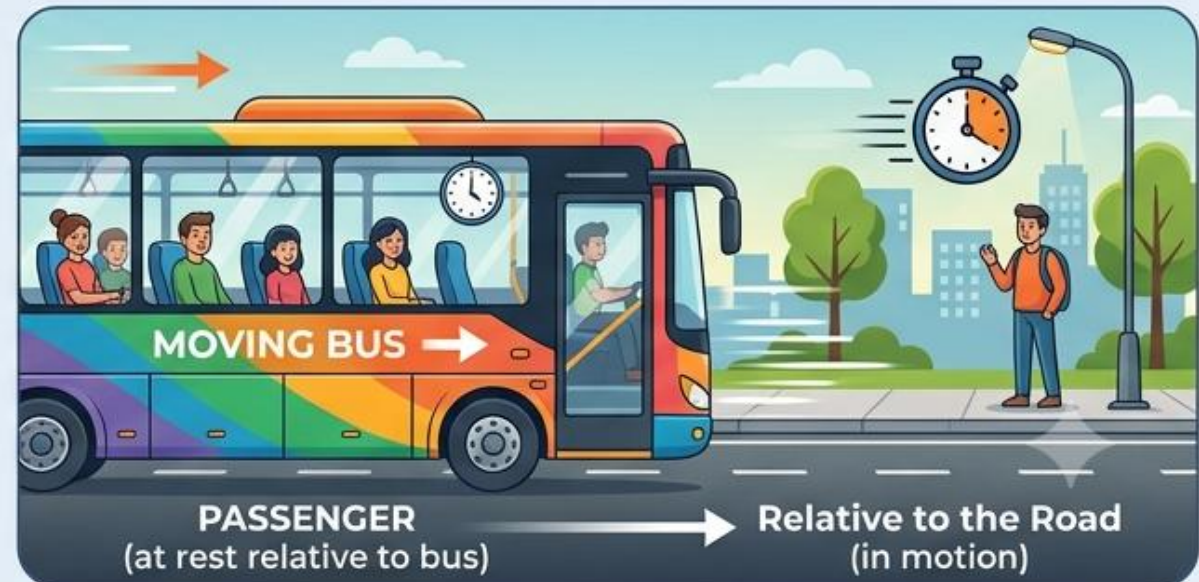
REFERENCE POINT

– a fixed point used for comparison.



TIME

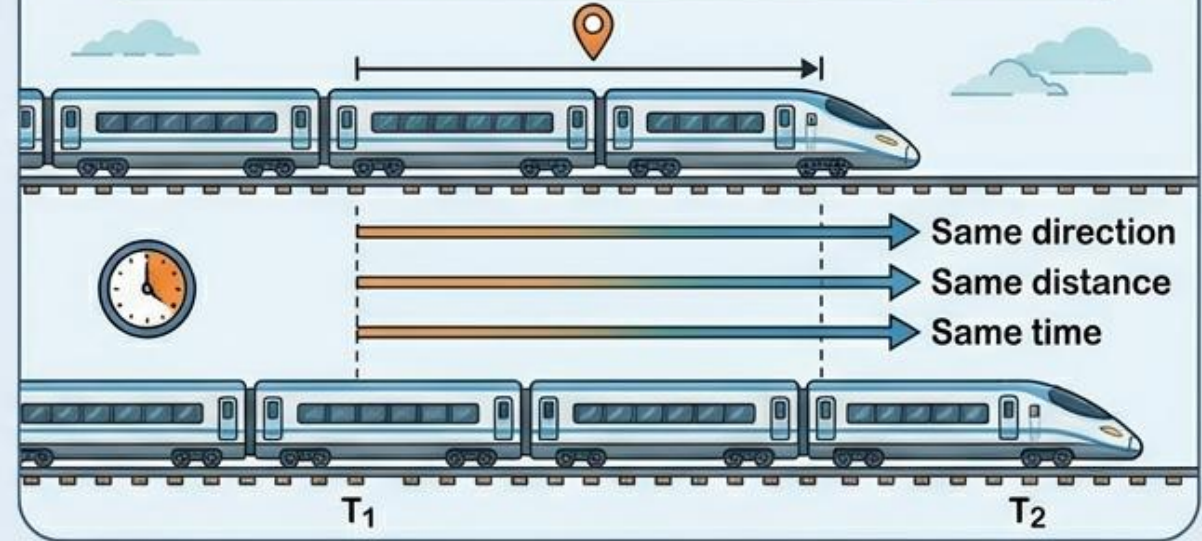
– the duration during which motion occurs.



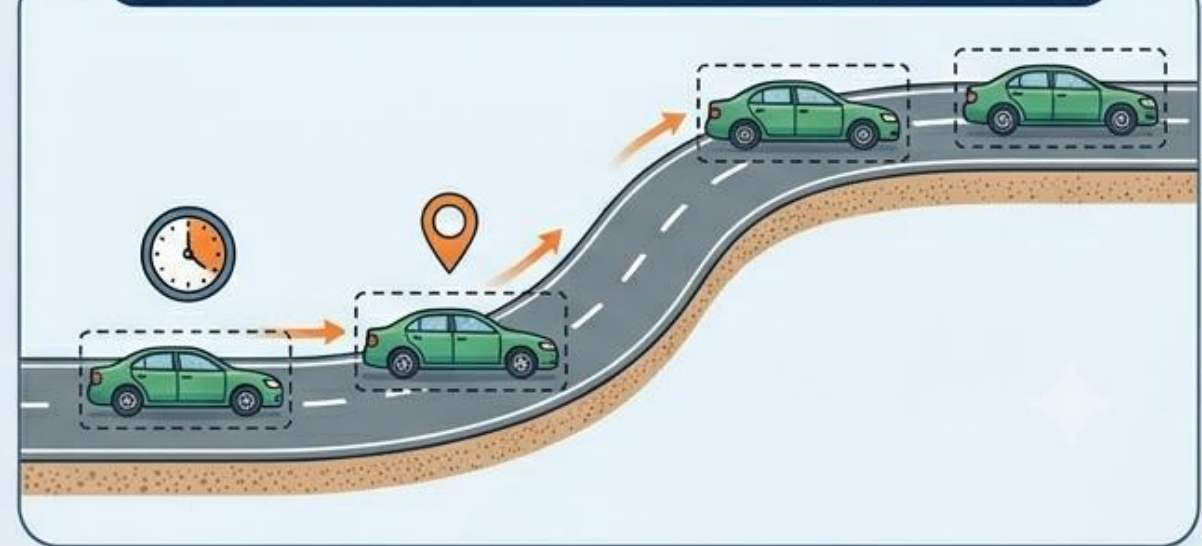
Translational Motion

- **Translational motion** is the movement of an object from one place to another without changing its orientation. All parts of the object move together in the same direction and cover the same distance at the same time.
- The motion can be along a straight path or a curved path.

Rectilinear Translational Motion (Straight Path)



Curvilinear Translational Motion (Curved Path)

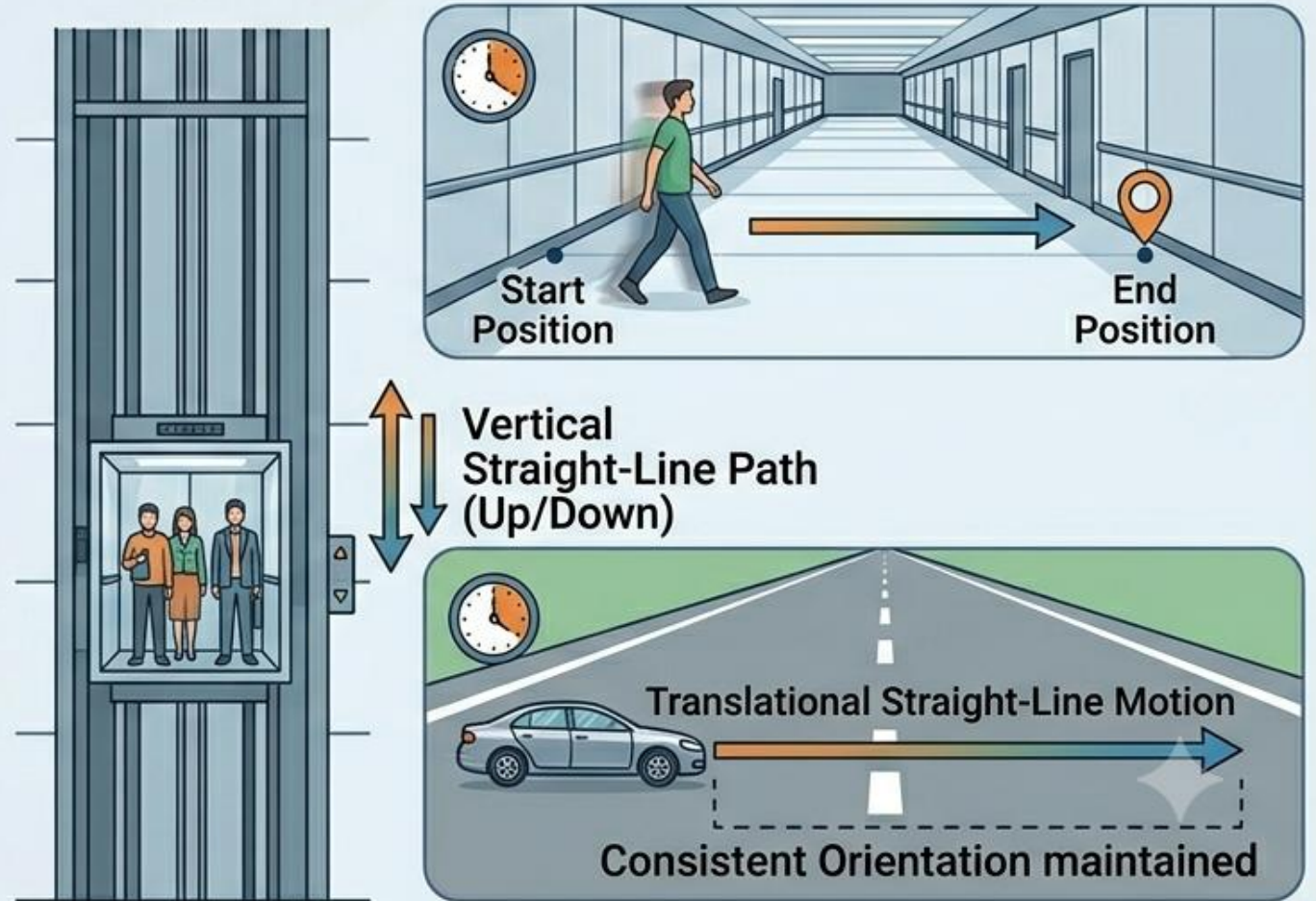


1. Rectilinear Motion

- This is translational motion along a straight-line path.

Examples:

- A person walking down a straight hallway
- An elevator moving up or down
- A car traveling on a straight road

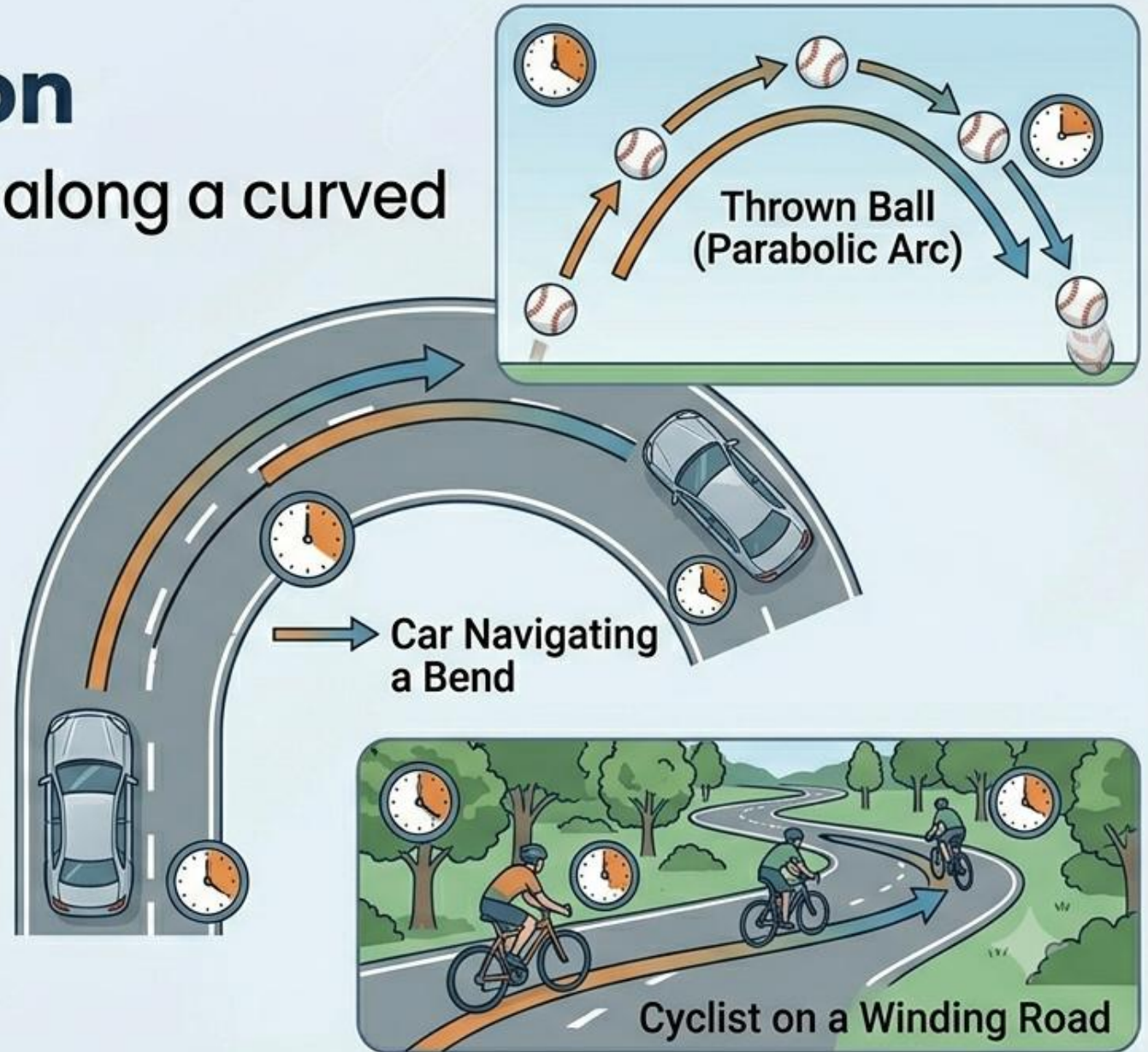


2. Curvilinear Motion

- This is translational motion along a curved path.

Examples:

- A thrown ball moving through the air
- A car turning around a bend
- A cyclist riding along a curved road

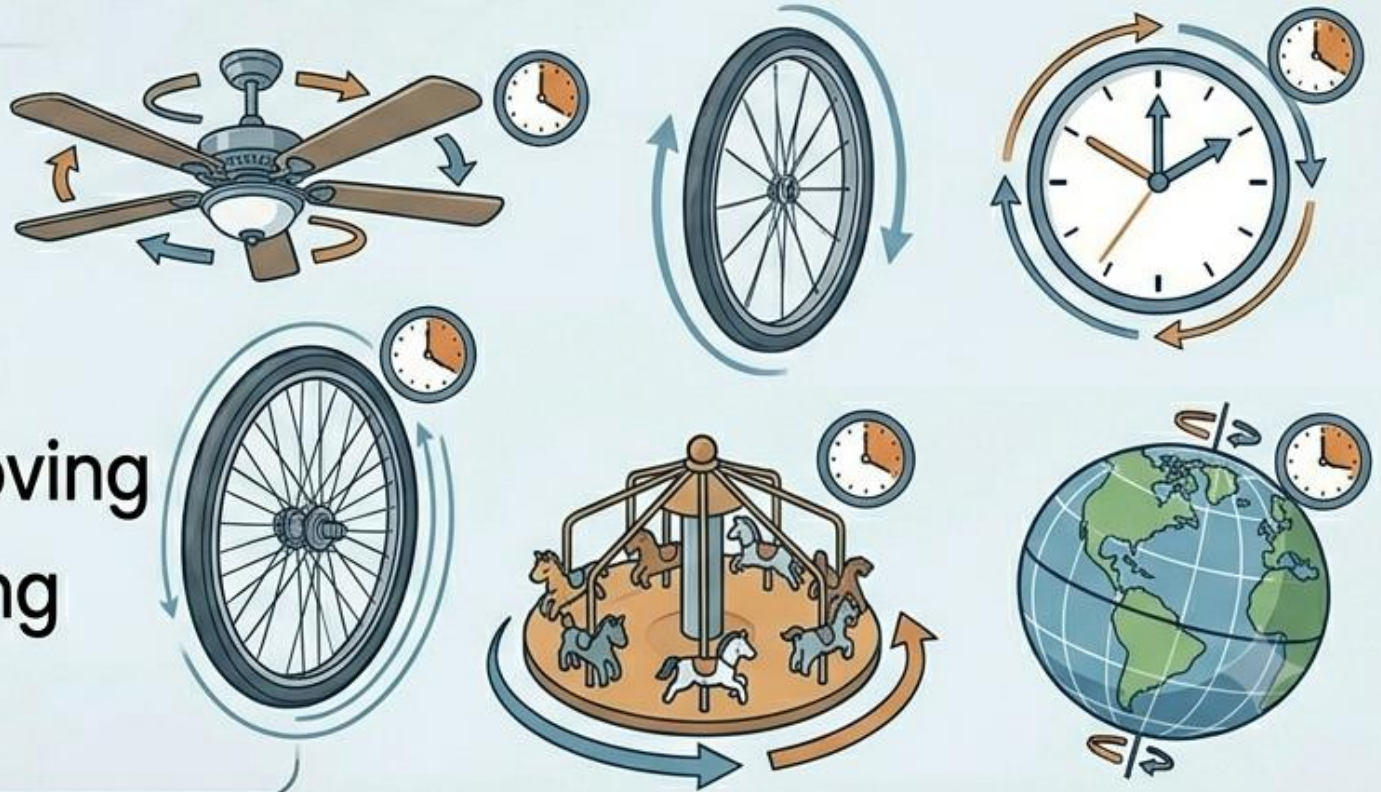


Rotational Motion

- Rotational motion is the motion of an object as it spins or turns about a fixed point or axis. As the object rotates, its parts move in circular paths around the axis of rotation.

Examples:

- A ceiling fan spinning
- A bicycle wheel rotating
- The hands of a clock moving
- A merry-go-round turning
- Earth rotating on its axis



TYPES OF MOTION

ROTATIONAL MOTION

SCIENCE

for Kids

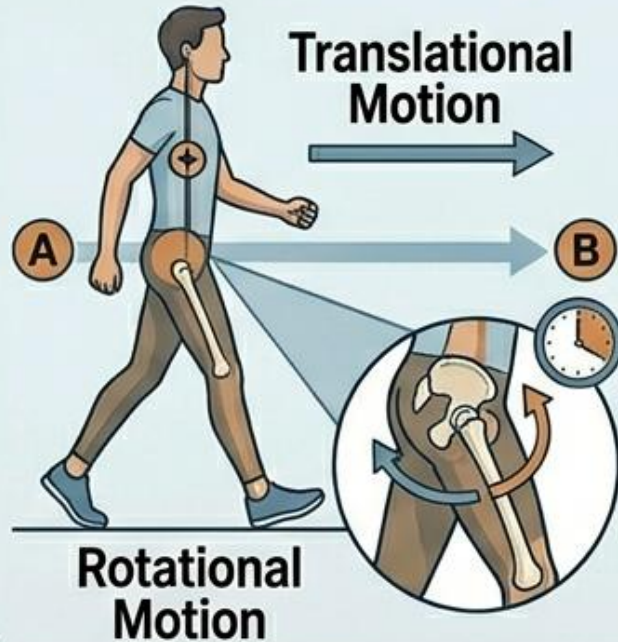


Applications of Translational and Rotational Motion

- Human movement often combines translational motion (moving from one place to another) and rotational motion (turning around a joint or axis).



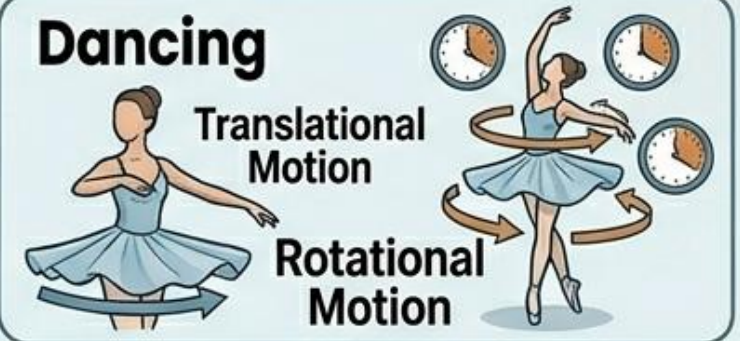
Walking/Running



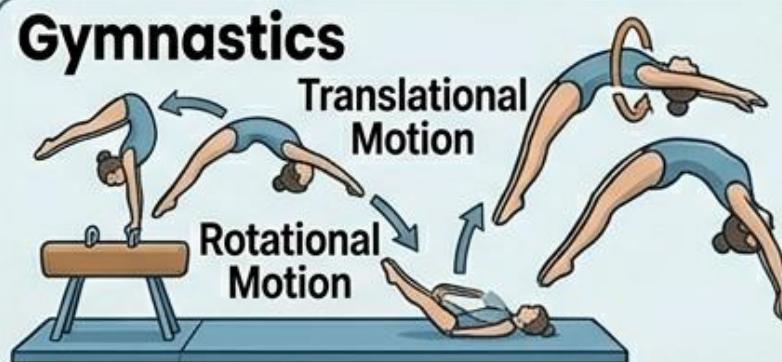
Dancing



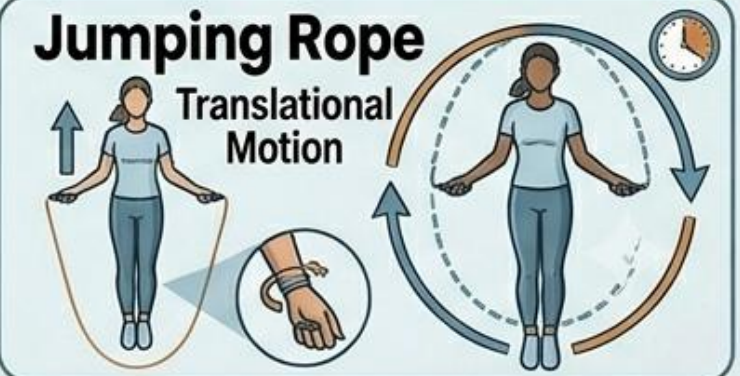
Dancing

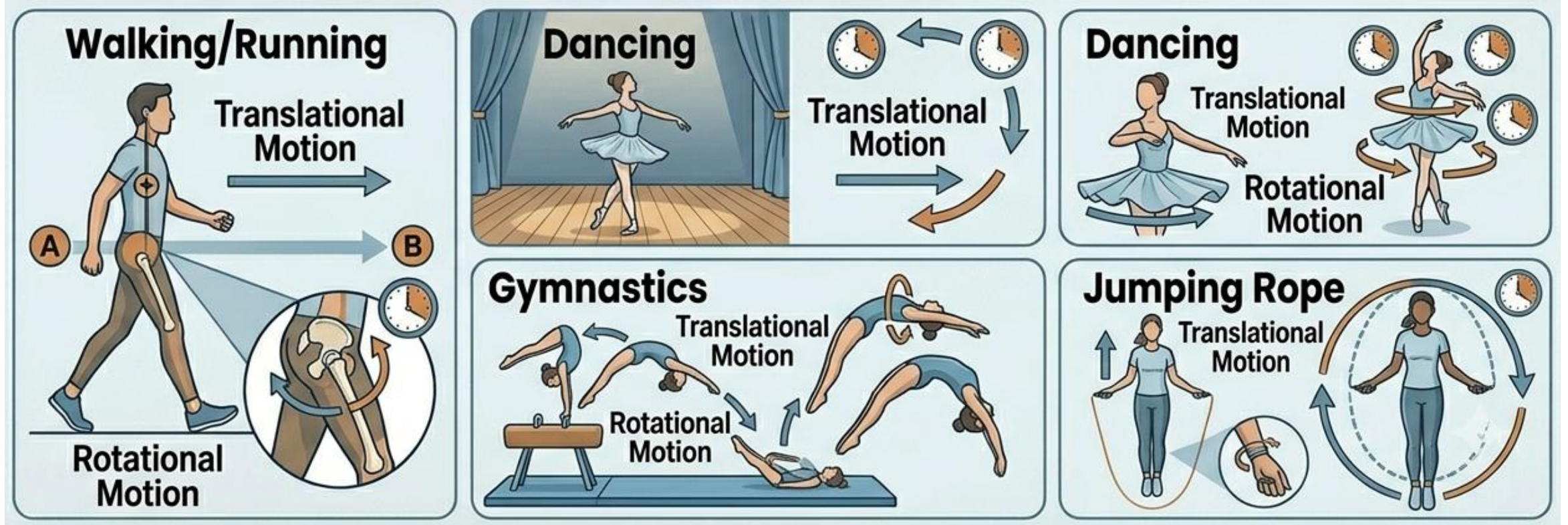


Gymnastics



Jumping Rope





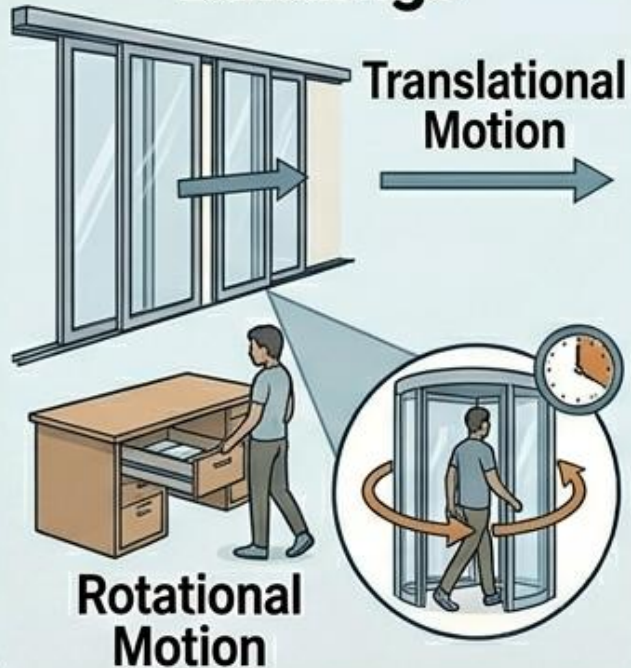
Activity	Translational Motion	Rotational Motion
Walking or Running	The body moves from one place to another.	The legs swing about the hip joints.
Dancing	The dancer moves across the floor.	The dancer spins or turns.
Gymnastics	The gymnast moves across a mat or through the air.	The body rotates during flips and turns.
Jumping Rope	The person moves up and down slightly.	The rope rotates around the body.

Ergonomic Designs

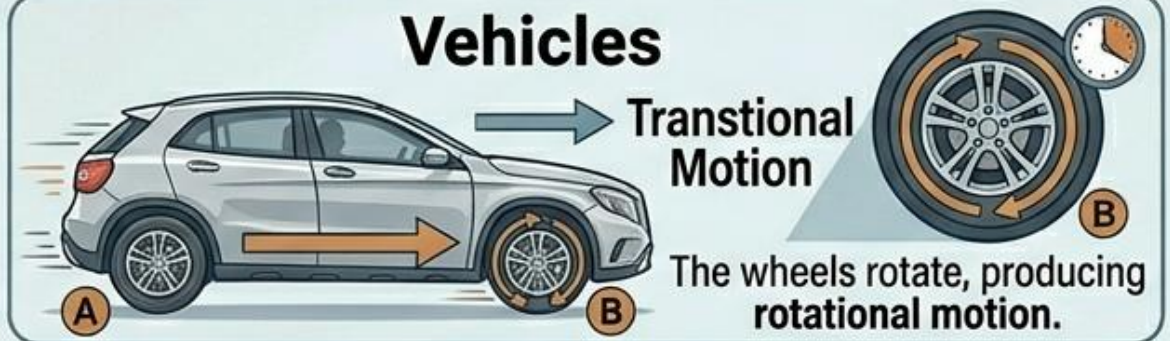
- Many everyday objects are designed using translational and rotational motion to make them easier, safer, and more comfortable to use.



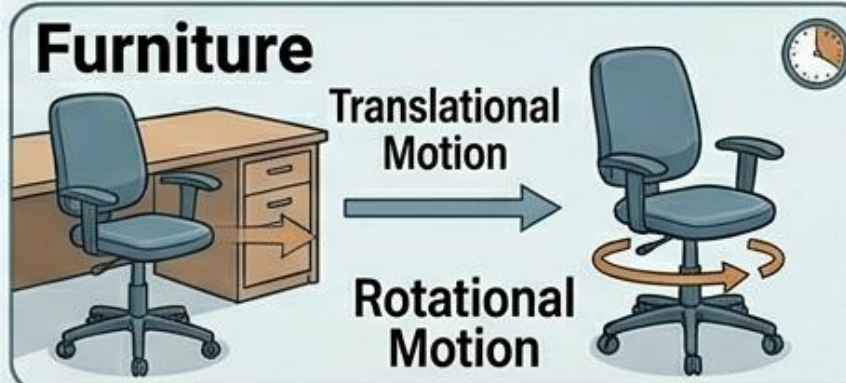
Buildings



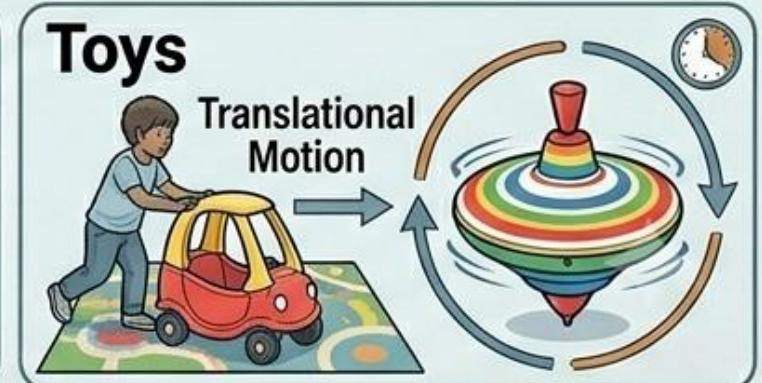
Vehicles

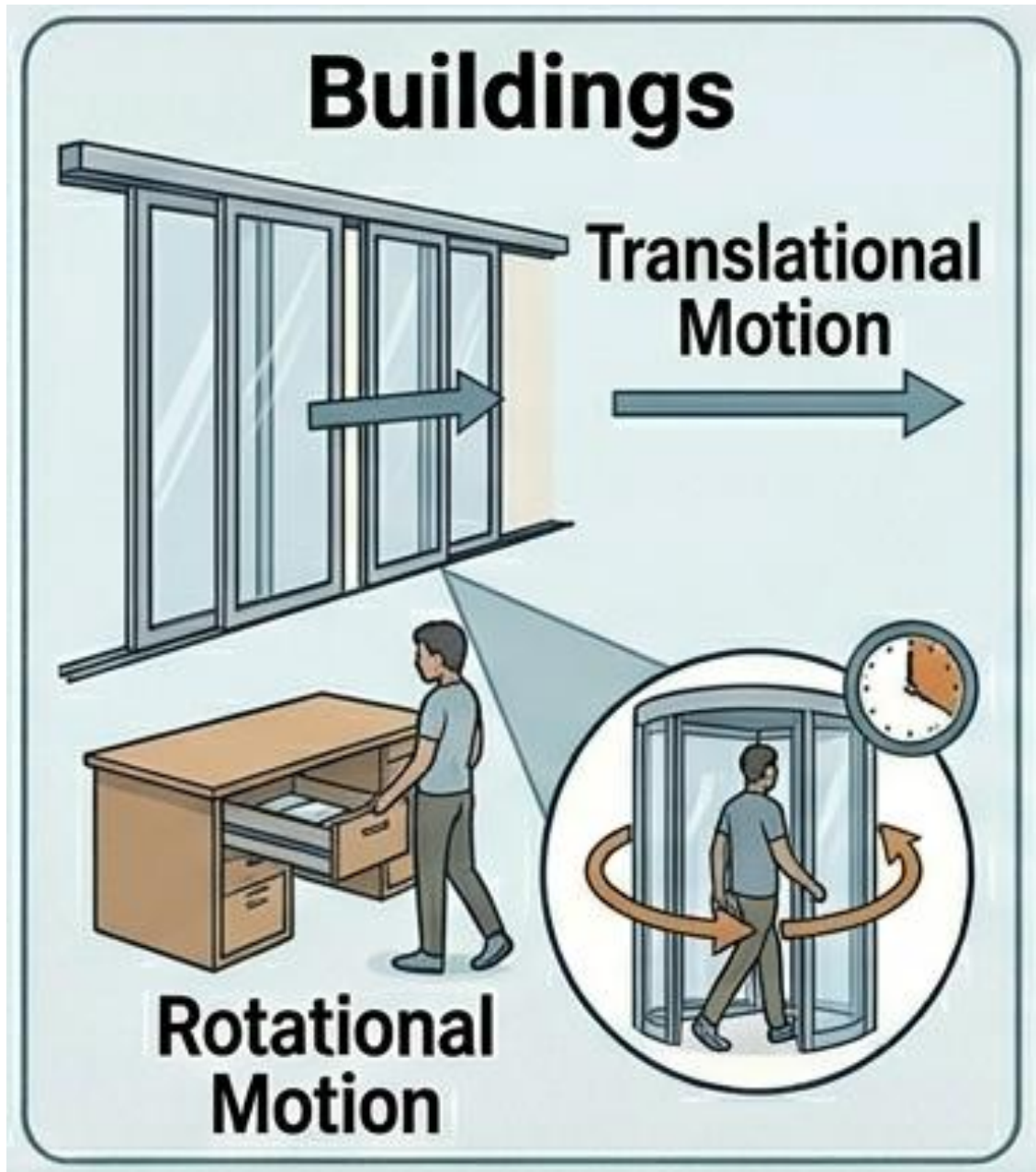


Furniture



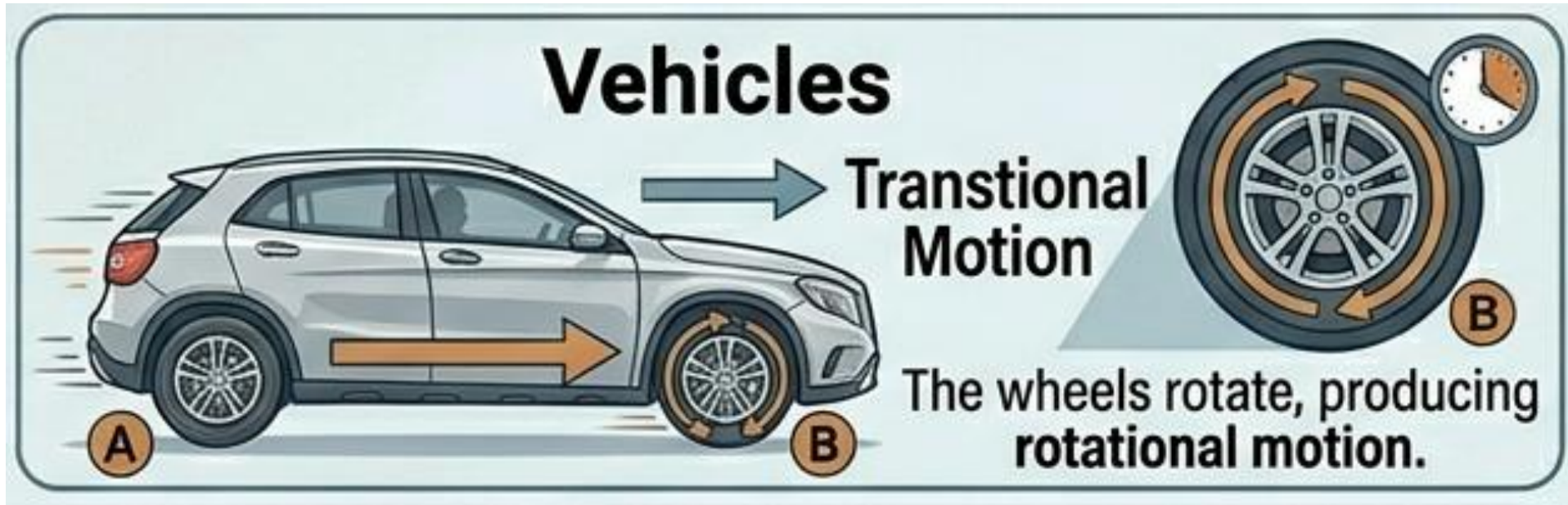
Toys





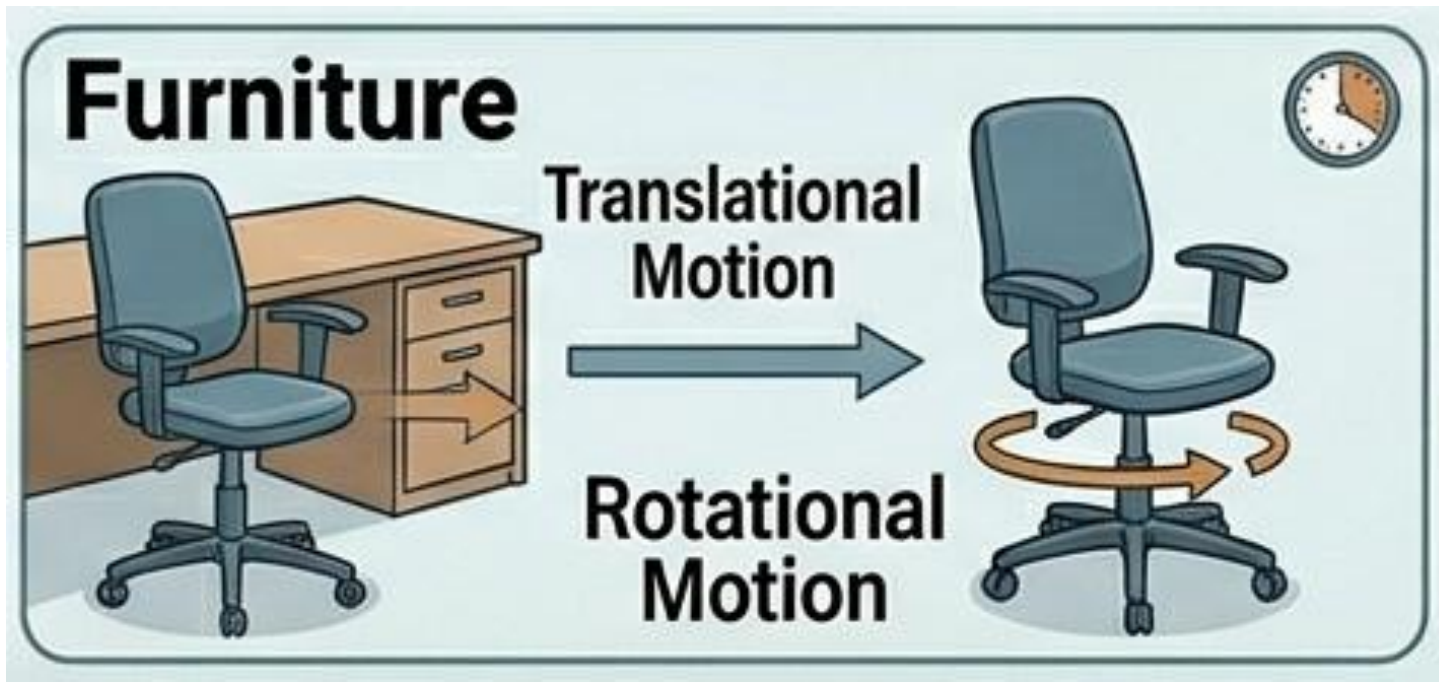
Buildings

- Sliding doors move in translational motion.
- Revolving doors move in rotational motion.



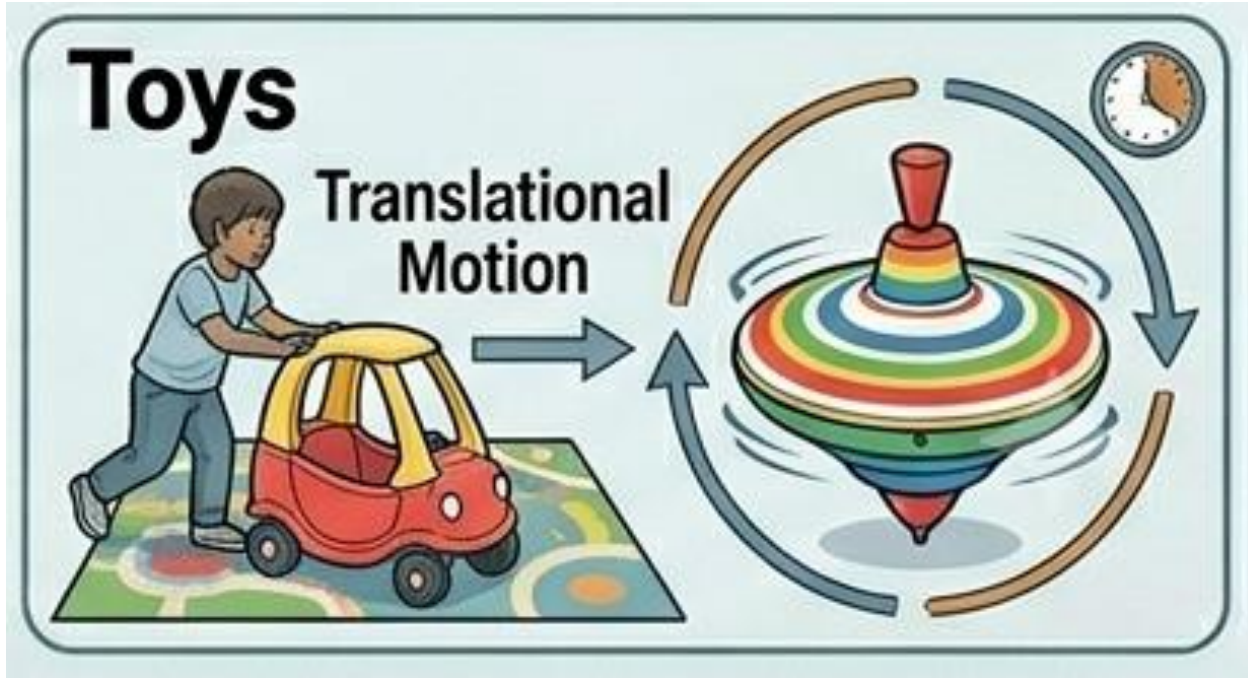
Vehicles

- A car moves forward through translational motion.
- The wheels rotate, producing rotational motion.



Furniture

- **A drawer slides in and out through translational motion.**
- **A swivel chair rotates around its central support.**



Toys

- A toy car moves across the floor through translational motion.
- A spinning top rotates around its axis.